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## Integrable Systems — Exercise Sheet 20

*Generalised hydrodynamics, and the quantum Newton's cradle*

**Variant: Questions only**

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Conventions, as in Lecture 20. Hard rods: length  $a$ , unit mass, rapidity  $\lambda$  the velocity,  $p_0 = \lambda$ ,  $\varepsilon_0 = \lambda^2/2$ ; a state is a density  $\rho(\lambda)$  with  $n = \int \rho$ ,  $J = \int \lambda \rho$ , and  $an < 1$ . Dressing, as in Lecture 17:  $f^{\text{dr}} = f + \frac{1}{2\pi} \int K(\cdot - \mu) \vartheta(\mu) f^{\text{dr}}(\mu) d\mu$  with  $\vartheta = \rho/\rho_t$  and  $2\pi\rho_t = (p'_0)^{\text{dr}}$ ;  $v^{\text{eff}} = (\varepsilon'_0)^{\text{dr}}/(k'_0)^{\text{dr}}$ ; generalised hydrodynamics  $\partial_t \rho + \partial_x(v^{\text{eff}} \rho) = 0$  for every  $\lambda$ . The Lieb–Liniger gas is in units  $\hbar = 2m = 1$ , with  $\varepsilon_0 = k^2$  and  $K(k) = 2c/(k^2 + c^2)$ .

**Exercise 1 (hard rods).** (i) From Lecture 20's rate equation  $v(\lambda) = \lambda + a \int \rho(\alpha) [v(\lambda) - v(\alpha)] d\alpha$ , show that  $\int \rho v = J$ , and deduce  $v^{\text{eff}} = (\lambda - aJ)/(1 - an)$ . (ii) Show that dressing with the constant kernel  $K = -a$  gives  $(1)^{\text{dr}} = 1 - an$  and  $(\lambda)^{\text{dr}} = \lambda - aJ$ , and so the same  $v^{\text{eff}}$ . (iii) Boost every rod by  $u$ . Show that  $v^{\text{eff}}$  is boosted by  $u$  too.

**Exercise 2 (the gas at large  $c$  is rods of negative length).** (i) Show that  $K(k) = 2/c + O(k^2/c^3)$ . (ii) Dress the Lieb–Liniger gas with the constant kernel  $K_0 = 2/c$  and show that  $v^{\text{eff}} = 2(k - aJ)/(1 - an)$  with  $a = -2/c$ : hard rods of negative length, as Lecture 17's large- $c$  limit says. (iii) Show that a local kernel  $K = \kappa \delta$  would multiply every  $f$  by the same factor wherever  $\kappa \vartheta \neq 2\pi$ , so that  $v^{\text{eff}}$  would not be dressed at all.

**Exercise 3 (which ratio is the velocity).** Lecture 20's generalised TBA is  $\eta(\lambda) = w(\lambda) - \frac{1}{2\pi} \int K(\lambda - \mu) \ln(1 + e^{-\eta(\mu)}) d\mu$ ,  $\vartheta = 1/(1 + e^\eta)$ . (i) Show that  $w = (\varepsilon_0 - \mu)/T$  gives Lecture 17's Yang–Yang equation, with  $\eta = \varepsilon/T$ . (ii) Show that  $\eta' = (w')^{\text{dr}}$ , assuming  $K \rightarrow 0$  at infinity and  $\ln(1 + e^{-\eta})$  bounded. (iii) For free fermions ( $K = 0$ ) with  $\varepsilon_0 = \lambda^2$  and  $w = \beta_2 \lambda^2 + \beta_4 \lambda^4$ , compare  $v^{\text{eff}}$  with  $\eta'/(2\pi\rho_t)$ .

**Exercise 4 (the cradle's free limits).** In the ideal and the impenetrable limits  $v^{\text{eff}} = 2\lambda$ . In a harmonic trap Lecture 20's characteristics are  $\dot{x} = 2\lambda$ ,  $\dot{\lambda} = -\partial_x V$ . (i) Why is  $V = \omega^2 x^2/4$  the harmonic trap of frequency  $\omega$  in these units? Solve the characteristics, and show that they rotate the plane  $(x, 2\lambda/\omega)$  rigidly with period  $2\pi/\omega$ . (ii) Deduce that the filling returns exactly after each period. Show that at a quarter period positions and rescaled rapidities exchange roles, so that the momentum distribution  $M_t(\lambda) = \int \rho dx$  is then  $M_{\pi/2\omega}(\lambda) = \frac{2}{\omega} n_0(-2\lambda/\omega)$ , with  $n_0 = \int \rho d\lambda$  at  $t = 0$ . Is  $M_t$  constant for every initial state?

**Exercise 5 (a lump of rods splitting).** (*Computational.*) Take hard rods with  $a = 1$  and the initial state  $\rho_0(\lambda, x) = n_0(x) p(\lambda)$ ,  $n_0(x) = 0.1(1 + 3e^{-(x/6)^2})$ , with  $p$  the normalised sum of two Gaussians of width 0.45 centred at  $\lambda = \pm 2$ : the state of Lecture 20's figure. The background of density 0.1 fills the whole line; the lump is the excess over it. (i) Solve generalised hydrodynamics by the exact method of Exercise 7\*. Plot  $n(x, t)$  at  $t = 0, 4, 8, 12$ , and the free ( $a = 0$ )  $n(x, 12)$ , and locate the right-moving peak of  $n(x, 12)$  for  $a = 1$  and for  $a = 0$ . (ii) Solve it again by an upwind scheme in flux form on a periodic grid. Check that the scheme conserves the rods of each rapidity slice, and compare the two solutions at  $t = 12$ . (iii) Why does the interacting lump run ahead?

**Exercise 6 (the Dictionary).** The course's closing outcome. For each Part I entry below, give its Part II counterpart, the lecture where it lives, and the type of the correspondence: **T** a

theorem of this course (an explicit proved map or  $\hbar \rightarrow 0$  limit), **A** a structural analogy, **C** cited. Poisson bracket;  $n$  integrals in involution; Poincaré sections; the linear action–angle system;  $\oint p \cdot dq = 2\pi\hbar n$ ; permuted momenta; the classical displacement; the Lax matrix; the Sklyanin bracket;  $r(\lambda)$ ;  $\{t(\lambda), t(\mu)\} = 0$ ;  $\text{tr } L^k$ ; the classical Yang–Baxter equation; the spectral curve; bands and gaps; the frequency  $\partial K/\partial I$ ; the measure  $dI d\varphi$ ; Calogero–Moser by reduction, and isochrony; the roots of  $\mathfrak{sl}_N$ ; elastic non-diffractive scattering; invariant tori; KAM.

**Exercise 7 ★.** *Hard rods are contracted free particles.* Let point particles stream freely in a coordinate  $y$ :  $\partial_t \hat{\rho} + \lambda \partial_y \hat{\rho} = 0$ , with  $\hat{\rho} \geq 0$  smooth, and  $\hat{n} = \int \hat{\rho} d\lambda$ ,  $\hat{j} = \int \lambda \hat{\rho} d\lambda$  finite and differentiable under the integral. No decay as  $|y| \rightarrow \infty$  is assumed. (i) Show that  $\hat{N}(y, t) = \int_0^y \hat{n}(y', t) dy' - \int_0^t \hat{j}(0, t') dt'$  has  $\partial_y \hat{N} = \hat{n}$  and  $\partial_t \hat{N} = -\hat{j}$ . Give each particle a length  $a > 0$  by the map  $x = y + a\hat{N}(y, t)$ , and let  $\rho(\lambda, x, t) dx = \hat{\rho}(\lambda, y, t) dy$ . (ii) Show that at each  $t$  the map is increasing and onto the line; that  $\rho = \hat{\rho}/(1 + a\hat{n})$ ,  $n = \hat{n}/(1 + a\hat{n})$ ,  $j = \hat{j}/(1 + a\hat{n})$  and  $1 - an = 1/(1 + a\hat{n})$ ; and that the point  $y$  at a fixed  $x$  moves as  $\partial_t y|_x = a\hat{j}/(1 + a\hat{n})$ . (iii) With  $\hat{C}(\lambda, y, t) = \int_0^y \hat{\rho}(\lambda, y', t) dy' - \lambda \int_0^t \hat{\rho}(\lambda, 0, t') dt'$  and  $C(\lambda, x, t) = \hat{C}(\lambda, y, t)$ , show that  $\partial_x C = \rho$  and  $\partial_t C|_x = -\rho v^{\text{eff}}$  with  $v^{\text{eff}} = (\lambda - aj)/(1 - an)$ , and deduce that  $\rho$  solves generalised hydrodynamics for hard rods.